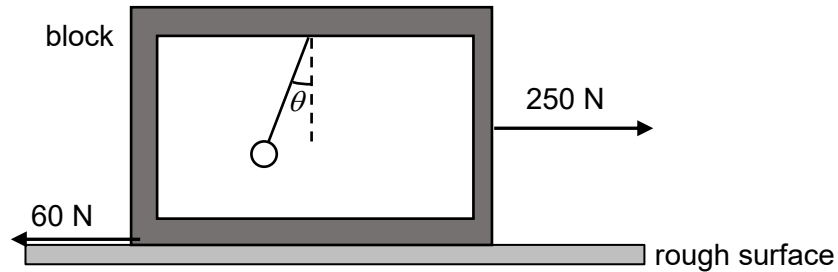


- 4 A block with a pendulum bob hanging lies on a rough surface. The total mass of the block and the pendulum bob is 100 kg. A force of 250 N is applied to the block and the pendulum bob makes an angle θ to the vertical axis.



What is the magnitude of θ if a constant frictional force of 60 N is acting on the block?

- A** 11° **B** 14° **C** 76° **D** 79°

Ans: A

Referring to FBD of whole system,

$$\Sigma F = ma$$

$$250 - 60 = 100a$$

$$a = 1.9 \text{ m s}^{-2}$$

Referring to FBD of bob,

$$\Sigma F_x = m_{\text{bob}} a$$

$$T \sin \theta = m_{\text{bob}} a \text{ ---- (1)}$$

$$\Sigma F_y = 0$$

$$T \cos \theta = m_{\text{bob}} g \text{ ---- (2)}$$

$$(1)/(2): \tan \theta = a / g = 1.9 / 9.81$$

$$\theta = 11^\circ$$

- 5 Which one of the following pair of forces is **not** an example of Newton's third law action and